

Experiment ‘p_1_4_account_balance’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_4_account_balance

Natural language statement: null

Method signature: p_1_4_account_balance() returns (first: real, second: real, third: real)

Ensures

- `first == 1000.0 + 1000.0 * 0.05`
- `second == first + first * 0.05`
- `third == second + second * 0.05`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_4_account_balance()` returns (first: real, second: real, third: real)

The method should respect the following contract:

ensures first == 1000.0 + 1000.0 * 0.05, ensures second == first + first * 0.05, ensures
 ↪ third == second + second * 0.05

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2DaIcCkQuoyuMEiUfhnDvdG1w3w

Model: gpt-5.5-2026-04-23

Created at: 1780578970

Finish reason: stop

Usage: 328 tokens in, and 185 tokens out

```
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Final Program

```
method p_1_4_account_balance() returns (first: real, second: real, third: real)
  ensures first == 1000.0 + 1000.0 * 0.05
  ensures second == first + first * 0.05
  ensures third == second + second * 0.05
{
  first := 1000.0 + 1000.0 * 0.05;
  second := first + first * 0.05;
  third := second + second * 0.05;
}
```

Total Token Usage

Input tokens: 328

Output tokens: 185

Reasoning tokens: 118

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780578969282, ended at 1780579010078, lasting 40796ms (40.80 seconds)

Iteration #1 started at 1780578969283, ended at 1780579010078, lasting 40795ms (40.80 seconds)